🎓IIT MADRAS | BS DEGREE PROGRAMME

Course-Aware Personalized Learning Companion

MILESTONE 5 EVALUATION REPORT: RAG Pipeline Completion, Learner Profiling & Personalized Quiz Generation

GROUP 3 PROJECT TEAM MEMBERS

Mayank Singh	Ali Jawad	Sachi Dhaturaha	Aaryan P. Maurya	Jibin V Mathews
23f1000598	22f3001825	21f1000471	22f1000559	21f1001895
@Mayank8IITM	@22f3001825	@21f1000471	@AryanPratap455	@21f1001895

Executive Summary & Core Objectives

OVERVIEW

Primary Objective

Complete Milestone 5 evaluation of the RAG-based learning assistant featuring deep integration of learner profiling, personalized quiz synthesis, knowledge gap detection, and comprehensive system testing.

Core System Deliverables

- FastAPI Endpoints Exposure & Integration
- Learner Profile Implementation with Mastery Tracking
- Topic Taxonomy Design covering 40+ Course Topics
- Interactive Frontend Pages (Chat, Quiz, Progress)

Functional Capabilities

- **Knowledge Gap Detection:** Multi-dimensional scoring derived from learner interactions.
- **Quiz Generation Engine:** Fully grounded MCQ synthesis with source citations.
- **Conversation Memory:** Preserved contextual chat sessions.
- **Qdrant Cloud Index:** Verified vector search across 9,427 transcripts.

Status: Full Milestone 5 Operational System Delivered & Validated.

Key Breakthrough Findings

BENCHMARKS

1.000	0.93	1.00	0.92
MRR @ 5	PRECISION @ 5	RECALL @ 5	FAITHFULNESS SCORE

Optimal Architecture

Hybrid Dense + Sparse + Cross-Encoder Reranker achieved perfect Mean Reciprocal Rank (MRR@5: 1.000) on benchmark query sets.

Generation Fidelity

Achieved 0.92 Faithfulness and 1.00 Answer Relevance, ensuring responses are accurately grounded in course materials.

Quiz Alignment

100% topic alignment and answerability for synthesized MCQs with realistic distractors (80%) and zero hallucination.

Project Scope & Environment

INFRASTRUCTURE

Evaluation Strategy

- Retrieval accuracy measurement on benchmark queries
- Answer faithfulness and context relevance scoring
- Learner profile tracking and gap detection accuracy
- Personalized quiz quality and distractor assessment
- System stability and out-of-scope query rejection

Technical Environment

Component	Specification
Operating System	Windows 11 (Build 10.0.26200)
Runtime / Framework	Python v3.10+, LangChain, FastAPI
Storage / Vector DB	SQLite (ORM) & Qdrant Cloud
Frontend Stack	React + Vite + Tailwind CSS
LLM Inference	Groq (Llama-3.3) & Google Gemini

System Architecture & Pipeline

ENGINEERING

RAG Pipeline

Hybrid search combining BAAI/bge-small dense vectors and BM25 sparse index, refined by Cross-Encoder reranking and evaluated via LLM-as-Judge.

FastAPI Backend

RESTful API endpoints facilitating interactive chat, learner profile state management, gap analytics, and dynamic quiz generation.

Database Layer

SQLite relational storage via SQLAlchemy ORM tracking student identity, chat session history, topic mastery scores, and quiz results.

Vector DB

9,427 indexed chunks stored in Qdrant Cloud enriched with structured metadata, temporal markers, and topic tags.

Content Pipeline

Structured chunking from Weeks 1–12 lecture transcripts, incorporating metadata injection and verified hierarchy.

React Frontend

Interactive single-page application built with Vite and Tailwind featuring Chat UI, Quiz module, and Real-time Progress Dashboard.

Quantitative Performance Results

EVALUATION

Metric Name	Score	Target Threshold	Interpretation & Impact
Precision@5	0.93	≥ 0.80	Exceptional relevance of top-5 retrieved context chunks
Recall@5	1.00	≥ 0.90	Perfect retrieval capture across benchmark query set
MRR@5	1.00	≥ 0.85	Optimal candidate retrieved consistently at position 1
Faithfulness	0.92	≥ 0.85	High groundedness; near-zero hallucination rate
Answer Relevance	1.00	≥ 0.90	Generated answers perfectly match user query intent
Quiz Topic Alignment	100%	100%	Generated test questions target specified weak topics
Out-of-Scope Rejection	100%	100%	Robust guardrail blocking irrelevant non-course queries

Data Infrastructure & Vector Index

KNOWLEDGE BASE

Corpus & Sources

- **Lecture Transcripts:** Complete Weeks 1–12 content with temporal timestamp markers.
- **Course FAQs:** Cleaned official instructor Q&A pairs.
- **Previous Year Papers:** Standardized practice questions and solutions.
- **Instructor Materials:** Slides, notes, and supplementary readings.

Infrastructure Statistics

Total Text Chunks	9,427 indexed chunks
Topic Taxonomy	40+ defined course topics
Average Chunk Size	200 – 300 tokens
Dense Embedding	BAAI/bge-small-en-v1.5
Sparse Index	BM25 Lexical Search
Topic Coverage	100% Weeks 1–12 Coverage

Personalization Engine & Quiz Synthesis

ADAPTIVE AI

Personalization Capabilities

- **Learner Profiling:** Tracks topic-wise mastery levels across all interactions.
- **Automatic Gap Detection:** Multi-dimensional scoring isolates weak concepts.
- **Dynamic Prioritization:** Automatically adjusts study plans based on quiz history.
- **Targeted Intervention:** Confidence-based routing triggers tailored recommendations.

Quiz Generation Engine

- **Topic-Targeted MCQs:** Synthesizes source-cited questions specifically targeting knowledge gaps.
- **Plausible Distractors:** Generates 4 distinct options with realistic incorrect choices (80% quality score).
- **Full Grounding:** Every synthesized question is tied to verified transcript source chunks.
- **Real-time Evaluation:** LLM-as-Judge framework validates difficulty and answerability.

Qualitative Case Studies & Safety

VALIDATION

Conceptual Query Handling

Query: "Explain overfitting vs underfitting in decision trees"

Outcome: Successfully retrieved relevant Weeks 3–4 chunks; generated a clear comparison table with precise video timestamps.

Guardrail Protection

Query: "What is quantum machine learning?"

Outcome: Correctly detected query as out-of-scope; returned a polite rejection preventing hallucination.

Formula Retrieval

Query: "What is the formula for Information Gain?"

Outcome: Retrieved exact mathematical definitions and explained the entropy reduction relationship accurately.

Weak Area Quiz Targeting

Target: "Ensemble Methods (Identified Weak Area)"

Outcome: Generated grounded MCQ on variance reduction with plausible distractors and transcript citations.

Technical Implementation Details

ARCHITECTURE

Component	Implementation Details	Technologies Used
RAG Pipeline	Hybrid search, Cross-Encoder reranking, LLM-as-Judge evaluation	LangChain, FastEmbed, Groq
Learner Profiles	Identity management, session history, mastery levels, progress tracking	SQLAlchemy, SQLite
Gap Detection	Multi-dimensional scoring algorithm derived from quiz attempts and chat history	Python algorithms, pandas
Quiz Engine	MCQ generation strictly grounded in course transcript materials	Groq llama-3.3, Prompting
Frontend Interface	Interactive Chat interface, Quiz page, and Progress Dashboard	React, Vite, Tailwind CSS
API Services	FastAPI REST service layer for chat, profiles, quizzes, and recommendations	FastAPI, Pydantic

Individual Team Contributions

TEAMWORK

Team Member	Roll No.	Key Contributions	Sign
Mayank Singh	23f1000598	Learner profiles, gap detection algorithms, recommendation logic, timestamp integration	[M.S]
Ali Jawad	22f3001825	FastAPI architecture, schema design, topic taxonomy, interactive frontend pages	[A.J]
Sachi Dhaturaha	21f1000471	Milestone presentation, evaluation report, technical documentation, synthesis	[S.D]
Jibin V Mathews	21f1001895	Quiz generation engine, system evaluation, conversation memory, QI module testing	[J.V.M]
Aaryan P. Maurya	22f1000559	Transcript processing, Qdrant index updates, metadata verification, team coordination	[A.P.M]

Team Consent Statement: All team members have thoroughly reviewed and consented to the contributions documented for Milestone 5.